

UNIVERSIDAD COMPLUTENSE DE MADRID

FACULTAD DE CIENCIAS FÍSICAS

Departamento de Física de la Tierra y Astrofísica

Group of Extragalactic Astrophysics and Astronomical
Instrumentation



Dwarfs4MOSAIC Repository

QUICK START GUIDE

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About this guide

Institution: Universidad Complutense de Madrid
Author: Noelia Grande Zafrilla
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Contact: Principal Investigator of Dwarfs4MOSAIC
<https://guaix.ucm.es/dwarfs4mosaic>

Intended Audience

[Part I](#) of the guide is designed for researchers and users without administrative privileges. It explains how to navigate and use the **dwarfs4MOSAIC repository** platform as a non-administrative user, providing step-by-step instructions.

[Part II](#) is intended for the **Administrator** user, with administrative privileges, and describes exclusively the platform's administrative functionalities, with step-by-step guidance.

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User's Guide

1. Accessing the Platform

To start using dwarfs4MOSAIC repository, open your preferred web browser and navigate to the project URL, <https://guaix.ucm.es/dwarfs4mosaic-data>. It is recommended to use modern browsers such as *Chrome*, *Firefox*, or *Edge* for optimal compatibility with all platform features. Once the page loads, you will reach the welcome page.



Figure I.1. Dwarfs4MOSAIC welcome page.

Click the **LOGIN** option at the top-right corner of the page to access the login form and enter your assigned username and password in the corresponding fields.

Figure I.2. Login form.

If you forget your password or experience login issues, please contact *Principal Investigator of Dwarfs4MOSAIC* at <https://guaix.ucm.es/dwarfs4mosaic> to request a new password.

2. User Options

After logging in, you will be directed to the *Home* page, which displays a welcome message along with several navigation options, providing quick access to the features most relevant to daily tasks.

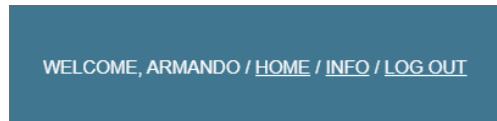
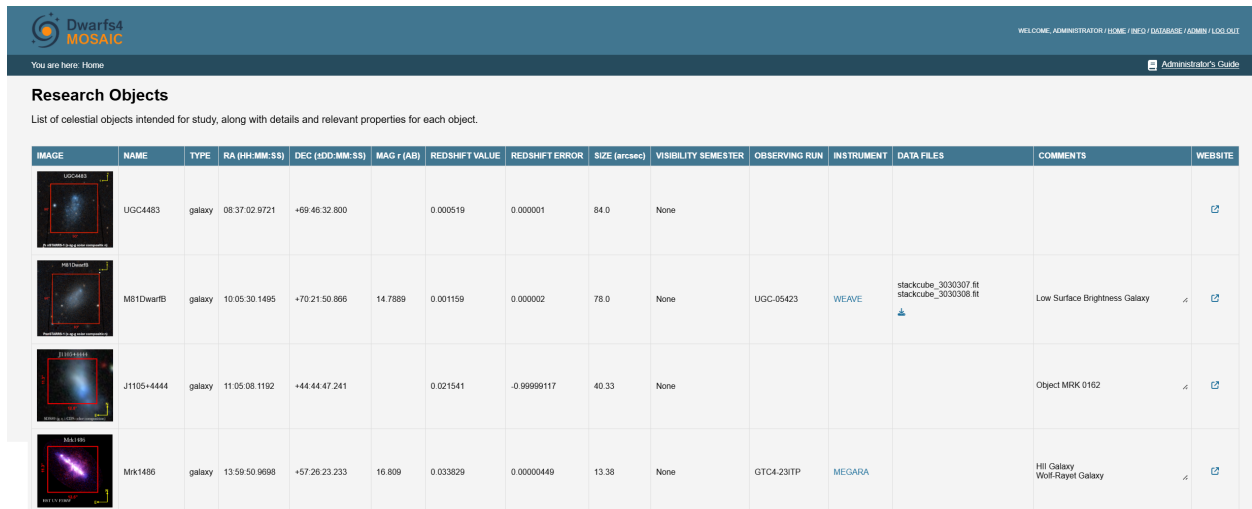


Figure I.3. Navigation links for a generic user.

- **Home:** redirects to the main page (see [Chapter 3. Navigating the Home Page](#)).
- **Info:** access to platform information (see [Chapter 5. Platform Information](#)).
- **Log out:** option to manually log out of the platform (see [Chapter 6. Logging Out](#)).

3. Navigating the Home Page

The *Home* page shows a table with the targets you have permission to view. Targets are displayed in ascending order of right ascension.



The screenshot shows the 'Dwarf4 MOSAIC' web interface. At the top, there's a navigation bar with links like 'WELCOME, ADMINISTRATOR / HOME / FIELD / DATABASE / ADMIN / LOG OUT' and a 'You are here: Home' breadcrumb. Below the header, the 'Research Objects' section is titled, followed by a subtitle: 'List of celestial objects intended for study, along with details and relevant properties for each object.' The main content is a table with 14 columns: IMAGE, NAME, TYPE, RA (HH:MM:SS), DEC (DD:MM:SS), MAG r (AB), REDSHIFT VALUE, REDSHIFT ERROR, SIZE (arcsec), VISIBILITY SEMESTER, OBSERVING RUN, INSTRUMENT, DATA FILES, COMMENTS, and WEBSITE. Four objects are listed: UGC4483, M81DwarfB, J1105+4444, and Mrk1486. Each row includes a thumbnail image, the object name, type (all are 'galaxy'), coordinates, magnitude, redshift, size, visibility semester, observing run, instrument, data files, comments, and a website link.

IMAGE	NAME	TYPE	RA (HH:MM:SS)	DEC (DD:MM:SS)	MAG r (AB)	REDSHIFT VALUE	REDSHIFT ERROR	SIZE (arcsec)	VISIBILITY SEMESTER	OBSERVING RUN	INSTRUMENT	DATA FILES	COMMENTS	WEBSITE
	UGC4483	galaxy	08:37:02.9721	+69:46:32.800		0.000519	0.000001	84.0	None					Website
	M81DwarfB	galaxy	10:05:30.1495	+70:21:50.866	14.7889	0.001159	0.000002	78.0	None	UGC-05423	WEAVE	stackcube_3030307 fit stackcube_3030308 fit	Low Surface Brightness Galaxy	Website
	J1105+4444	galaxy	11:05:08.1192	+44:44:47.241		0.021541	-0.99999117	40.33	None				Object MRK 0162	Website
	Mrk1486	galaxy	13:59:50.9698	+57:26:23.233	16.809	0.033829	0.00000449	13.38	None	GT44-231TP	MEGARA		HII Galaxy Wolf-Rayet Galaxy	Website

Figure I.4. Home page.

Each target is displayed with several fields that provide key information at a glance:

Image: target's associated image, if available.

Name: unique identifier of the target.

Type: category of the astronomical object.

Right Ascension: celestial coordinate along the equatorial plane [hh:mm:ss].

Declination: celestial coordinate perpendicular to the equatorial plane [dd:mm:ss].

Apparent Magnitude: brightness of the target as observed from Earth.

Redshift (z): measured redshift of the object.

Angular Size: apparent size of the target in the sky.

Visibility Semester: semester when the target is visible.

Observing Run: name of the observing run(s) the target belongs to.

Instrument: instrument used to acquire data for the target.

Data Files: list of associated files available for download (see [Chapter 4. Downloading Files](#)).

Comments: additional information.

Website: link to an external astronomical database or information page related to the target.

4. Downloading Files

On the *Home* page, locate the target of interest. In the **Data Files** column, you will see the available files along with the download icon . Clicking this icon opens the file download view.

Target: NGC 6822
Available datafiles for downloading. Please select files to download.

☒	FILE NAME	SIZE
☒	final_cube_GTC4-23ITP_OB0006.fit	5 bytes
☒	final_cube_GTC4-23ITP_OB0009.fit	5 bytes

Download

Figure I.5. File download view.

This view displays a table containing all the files associated with the selected target and their respective sizes. You can select one, multiple, or all files at once. If only a single file is selected, it will be downloaded directly. If multiple files are selected, a *.zip* archive containing all selected files is generated. The download process is handled directly by your web browser.

5. Platform Information

This view provides relevant information about the platform. All information is kept up-to-date by the Administrator to ensure accuracy.

Platform Information

Features

Data Release 1

- MEGARA
- WEAVE

Data Release 1

MEGARA

MEGARA is an IFU spectrograph at 10.4m Gran Telescopio Canarias (GTC) which covers a field of $12.5 \times 11.3 \text{ arcsec}^2$ using 567 hexagonal spaxels of 0.62 arcsec in size plus 56 sky spaxels. Each galaxy has been observed using three different LR-VPHs ($R=5000-7000$): VPH405_LR (wavelength range: 3653-4386 Å), VPH480_LR (4332-5196 Å) and VPH675_LR (6094-7300 Å).

The MEGARA Data Reduction Pipeline generates a final product which consists of the compilation of 623 fibre spectra stacked in rows (called RSS, which stands for Row Stacked Spectra). This RSS is later converted into a cube, for which a pixel size of 0.3 arcsecs has been used. The provided cubes are flux calibrated and corrected from extinction, diffuse light and cosmic rays.

WEAVE

WEAVE is a new multi-object survey and multi-integral-field-unit spectrograph for the 4.2-m William Herschel Telescope (WHT), of which the LIFU (Large Integral Field Unit) mode has been used for the data in this release. The LIFU covers a field of view $90 \text{ arcsec} \times 78 \text{ arcsec}$, where the science array comprises 547 fibres (filling factor of 0.55), each having a diameter on the sky of 2.6 arcsec, plus 8 more sky bundles of fibres. The galaxies in DR1 have been observed with the Low-resolution VPHs ($R = 2500$) for both Red and Blue arms, whose wavelength ranges are 5790-9590 Å and 3660-6060 Å, respectively.

The observed raw data of WEAVE are passed to the Cambridge Astronomy Survey Unit (CASU) for automatic pipeline processing, and the final product of the reduction, referred to as Level 1, is what is provided in this release. Each observation of a galaxy produces two cubes called 'stackcube'—one red and one blue. Each cube combines the three dithers performed to account for the spacing between the fibers.

Figure I.6. Platform information view.

6. Logging Out

To securely exit the platform, click the **LOG OUT** option located at the top-right corner of the page. This will end your session and return you to the welcome page. Your session will also automatically expire after 15 minutes of inactivity, requiring you to log in again. This helps protect your account on shared computers.

 Always log out when you finish using the system, especially on shared or public computers, to protect your account and data.

7. Support

For any questions, technical issues, or to request a new password, please contact *Principal Investigator of Dwarfs4MOSAIC* at <https://guaix.ucm.es/dwarfs4mosaic>.

Response times may vary, and users are encouraged to provide a clear description of the issue.

 Always keep your login credentials secure and do not share them with others.

Administrator's Guide

1. Accessing the Platform

For general instructions on logging in and accessing the platform, the Administrator should refer to *Chapter 1. Accessing the Platform* in *Part I. User's Guide* .

2. Administrator Options

After logging in, you will be directed to the *Home* page, which displays a welcome message along with several navigation options, providing quick access to the features most relevant to daily tasks.

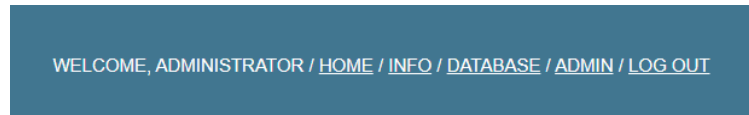


Figure II.1. Navigation links for the Administrator.

- **Home:** redirects to the main page (see [Chapter 3. Navigating the Home Page](#)).
- **Info:** access to platform information (see [Chapter 5. Platform Information](#)).
- **Database:** access to consult database tables (see [Chapter 6. Database Visualization](#)).
- **Admin:** access to administration interface to manage users, permissions, and tables (see [Chapter 7. Administration Panel](#)).
- **Log out:** option to manually log out of the platform (see [Chapter 8. Logging Out](#)).

3. Navigating the Home Page

For an overview of *Home* page navigation and target access, the Administrator should refer to *Chapter 3. Navigating the Home Page* in *Part I. User's Guide* .

4. Downloading Files

For general instructions on downloading files, the Administrator should refer to *Chapter 4. Downloading Files* in *Part I. User's Guide* .

5. Platform Information

This view provides relevant information about the platform and allows editing the page content. Next to the page title, there is a link (**Edit**) that scrolls to the editable section at the bottom of the page. This section contains an HTML editor widget where the **Administrator** can modify the page content directly using HTML.

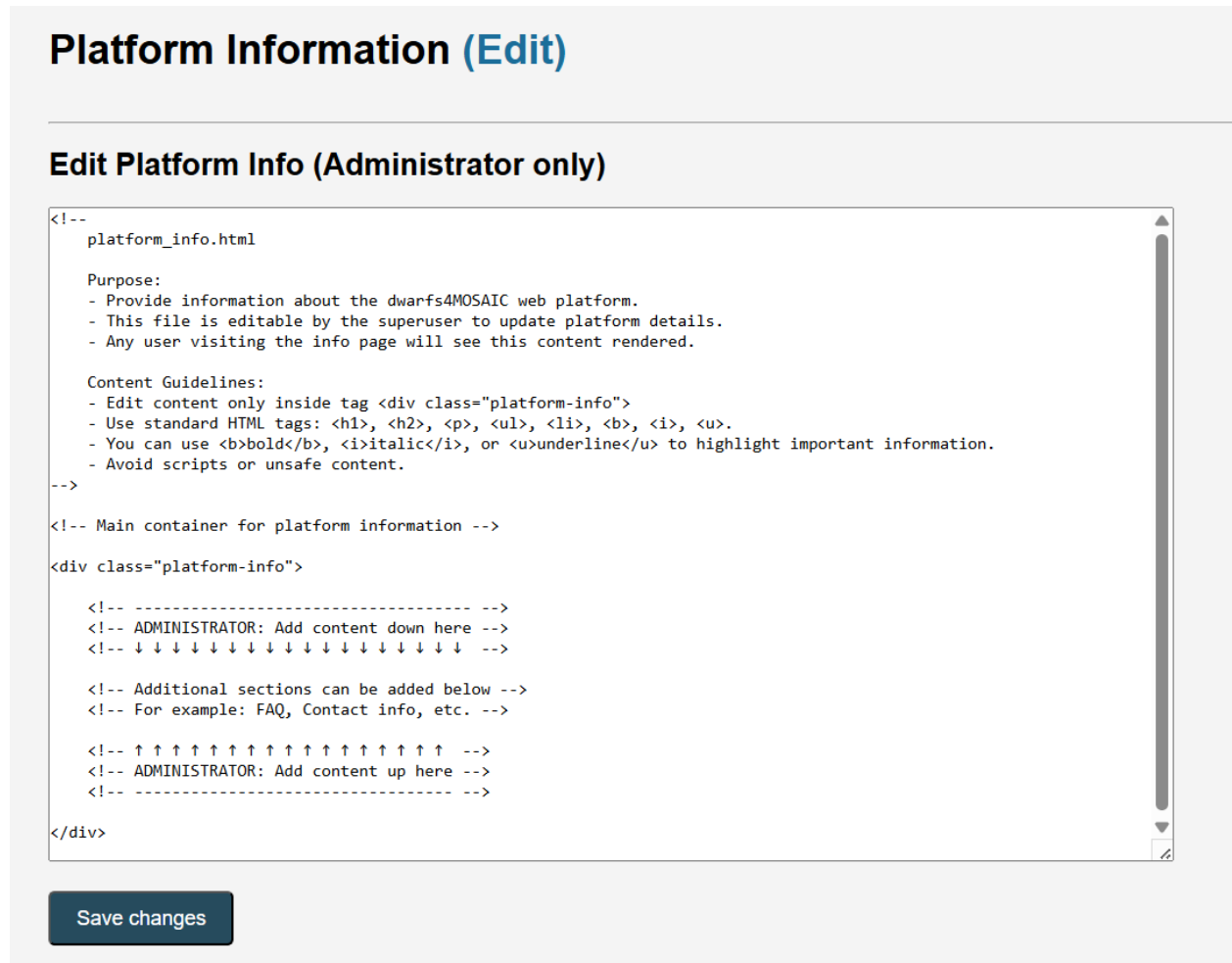


Figure II.2. Platform information view.

After making changes, the **Save changes** button stores the updated HTML on the server, and the page is immediately refreshed to display the new content. This feature ensures that the **Administrator** can update platform information efficiently while maintaining accuracy.

⚠ Changes made here will be immediately visible to all users. Ensure correctness before saving.

6. Database Visualization

This section introduces the interface for browsing the project’s database tables.

The diagram below shows the structure of the *dwarfs4MOSAIC* database, including the tables and the relationships between them. This overview helps understand how objects are interconnected.

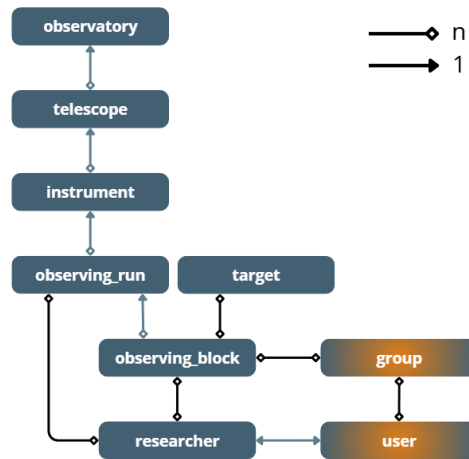


Figure II.3. Database structure and relationships between tables.

The platform allows the Administrator to explore the stored records, consult details quickly, and verify relationships between different objects. The feature ensures that the database content remains consistent with the project’s current state.

Database table query interface	
List of database tables displaying their fields and contents along with additional related information.	
TABLE NAME	
group	
instrument	
observatory	
observing_block	
observing_run	
researcher	
target	
telescope	

Figure II.4. Database tables interface.

The list of categories appears in alphabetical order; however, in the following sections, they are presented according to the order in which they are created, making it easier to understand the relationships between objects.

6.1 Groups

The *group* table contains information about the project groups used to organize members.

Each record represents a single group and shows its name, members (researchers) and authorized observing blocks for the researchers belonging to the group.

Groups table		
NAME	MEMBERS	ALLOWED OBSERVING BLOCKS
megara-group	agil acastillo ccabello mlara	GTC4-23ITP - MEGARA (2023-02-18)
weave-group		

Figure II.5. *Groups* table.

6.2 Researchers

The *researcher* table contains information about the project members.

Each record represents a single researcher and shows its full name, role (*Core Team* or *Collaborator*), the group it belongs to, whether it holds a PhD, affiliated institution, the observing runs it has participated in, specific denied observation blocks, email address and comments.

Researchers table							
NAME	ROLE	GROUP	PhD	INSTITUTION	OBSERVING RUNS	DENIED OBSERVING BLOCKS	EMAIL
África Castillo Morales	core_team	megara-group	✓	Universidad Complutense de Madrid, Spain	GTC4-23ITP		aca
Armando Gil de Paz	core_team	megara-group	✓	Universidad Complutense de Madrid, Spain	GTC4-23ITP		i
Cristina Cabello González	core_team	megara-group	✓	Universidad Complutense de Madrid, Spain			crist
Carolina Kehrig Martin dos Santos	core_team		✓	Instituto de Astrofísica de Andalucía, Spain	GTC4-23ITP	GTC4-23ITP - MEGARA (2023-02-18)	k
Christopher J. Conselice	core_team		✓	University of Nottingham, UK			conselic
Davor Krajinovic	core_team		✓	Institute Astrophysics Potsdam, Germany			dkr
Jorge Iglesias Páramo	core_team		✓	Instituto de Astrofísica de Andalucía, Spain			ji
José Manuel Vilchez Medina	core_team		✓	Instituto de Astrofísica de Andalucía, Spain	GTC4-23ITP	GTC4-23ITP - MEGARA (2023-02-18)	
Luca Costantin	core_team		✓	Centro de Astrobiología, Spain			lcostant
Matthew Hayes	core_team		✓	Stockholm University, Sweden			mattl
Maritza Lara López	core_team	megara-group	✓	Universidad Complutense de Madrid, Spain			m
Mathiew Puech	core_team		✓	Paris-Meudon Observatory, France			Mathie

Figure II.6. *Researchers* table.

6.3 Observatories

The *observatory* table contains information about the observatories associated with the project.

Each record represents a single observatory and shows its name, location, geographic longitude, geographic latitude, altitude above sea level and a link to a website with additional information.

Observatories table					
NAME	LOCATION	LONGITUDE (DD:MM:SS)	LATITUDE (DD:MM:SS)	ALTITUDE (m)	WEBSITE
AURA-O	Cerro Pachón (Chile)	70:44:01.11W	30:14:16.41N	2713.0	↗
Roque de los Muchachos	La Palma (Spain)	17:53:30W	28:45:22N	2396.0	↗

Figure II.7. *Observatories* table.

Clicking on an observatory name in the list a detailed page opens displaying all telescopes related to that observatory.

Observatory: Roque de los Muchachos					
Telescopes belonging to the observatory.					
TELESCOPE	DESCRIPTION	OWNER	STATUS	APERTURE (m)	WEBSITE
GTC	Gran Telescopio de Canarias	GRANTECAN S.A.	↗ operational	10.4	↗
INT	Isaac Newton Telescope	Instituto de Astrofísica de Canarias (IAC), Isaac Newton Group of Telescopes - La Palma, Nederlandse Organisatie voor Wetenschappelijk Onderzoek	↗ maintenance	2.5	↗
LT	Liverpool Telescope	Liverpool John Moores University	↗ operational	2.0	↗
WHT	William Herschel Telescope	Instituto de Astrofísica de Canarias (IAC)	↗ operational	4.2	↗

Figure II.8. Specific observatory information.

6.4 Telescopes

The *telescope* table contains information about the telescopes associated with each observatory.

Each record represents a single telescope and shows its name, description, observatory to which it belongs, institutional owner, operational status (*Unknown*, *Operational*, *Inoperative*, or *Under Maintenance*), aperture and a link to a website with additional information.

Telescopes table

NAME	DESCRIPTION	OBSERVATORY	OWNER	STATUS	APERTURE (m)	WEBSITE
GTC	Gran Telescopio de Canarias	Roque de los Muchachos	GRANTECAN S.A.	operational	10.4	↗
INT	Isaac Newton Telescope	Roque de los Muchachos	Instituto de Astrofísica de Canarias (IAC), Isaac Newton Group of Telescopes - La Palma, Nederlandse Organisatie voor Wetenschappelijk Onderzoek	maintenance	2.5	↗
LT	Liverpool Telescope	Roque de los Muchachos	Liverpool John Moores University	operational	2.0	↗
SOAR	Southern Astrophysical Research	AURA-O	The SOAR Consortium	operational	4.1	↗
WHT	William Herschel Telescope	Roque de los Muchachos	Instituto de Astrofísica de Canarias (IAC)	operational	4.2	↗

Figure II.9. Telescopes table.

Clicking on a telescope name in the list a detailed page opens showing all instruments related to that telescope.

Telescope: GTC (Gran Telescopio de Canarias)						
Instruments belonging to the telescope.						
INSTRUMENT	DESCRIPTION	FILTERS	CONFIGURATION	STATUS	WEBSITE	
MEGARA	Multi-Espectrógrafo en GTC de Alta Resolución para Astronomía	LR-U LR-B LR-R		inoperative	↗	

Figure II.10. Specific telescope information.

6.5 Instruments

The *instrument* table contains information about the instruments available for each telescope.

Each record represents a single instrument and shows its name, description, filters, configuration, telescope on which it is installed, operational status (*Unknown*, *Operational*, *Inoperative*, or *Under Maintenance*) and a link to a website with additional information.

Instruments table						
NAME	DESCRIPTION	FILTERS	CONFIGURATION	TELESCOPE	STATUS	WEBSITE
IO:O	Infrared-Optical suite of instruments: optical imaging component			LT	operational	↗
MEGARA	Multi-Espectrógrafo en GTC de Alta Resolución para Astronomía	LR-U LR-B LR-R		GTC	inoperative	↗
SIFS	SOAR Integral Field Spectrograph			SOAR	operational	↗
WEAVE	WHT Enhanced Area Velocity Explorer		LR-blue LR-red	WHT	operational	↗
WFC	Wide-Field Camera			INT	inoperative	↗

Figure II.11. Instruments table.

6.6 Observing Runs

The *observing_run* table contains information about the periods during which observations are scheduled for each instrument.

Each record represents a single observing run and shows its name, instrument used, description, start and end dates, and comments.

Observing Runs table					
NAME	INSTRUMENT	DESCRIPTION	START DATE	END DATE	COMMENTS
GTC4-23ITP	MEGARA	The observing run will be conducted at the 10.4m Gran Telescopio Canarias (GTC) in the framework of the International Time Programme 2023 (23ITP). It consists of a series of observing blocks targeting selected galaxies, each designed to deliver integral field spectroscopy datacubes for subsequent analysis.	Feb. 18, 2023	Feb. 19, 2023	The run was executed over several nights under photometric conditions with typical seeing below 1.0".

Figure II.12. *Observing Runs* table.

Clicking on an observing run name in the list a detailed page opens showing all researchers involved in the observing run as well as all the observing blocks included.

Observing Run: GTC4-23ITP

Researchers involved in the observing run

NAME	ROLE	PhD	INSTITUTION	EMAIL	COMMENTS
África Castillo Morales	core_team	✓	Universidad Complutense de Madrid, Spain	acasmor@ucm.es	
Armando Gil de Paz	core_team	✓	Universidad Complutense de Madrid, Spain	agil@ucm.es	
Carolina Kehrig Martin dos Santos	core_team	✓	Instituto de Astrofísica de Andalucía, Spain	kehrig@iaa.es	
José Manuel Vilchez Medina	core_team	✓	Instituto de Astrofísica de Andalucía, Spain	jvm@iaa.es	

Observing blocks included in the observing run

BLOCK NAME	DESCRIPTION	START TIME	END TIME	OBSERVATION MODE	FILTERS	CONFIGURATION	EXPOSURE TIME	SEEING	WEATHER CONDITIONS
GTC4-23ITP	NGC 6822, IFS with MEGARA (23ITP).	Feb. 18, 2023, midnight	12:10 a.m.	photometry	LR-U				

Figure II.13. Specific observing run information.

6.7 Observing Blocks

The *observing_block* table contains information about the specific set of scheduled observations for each observing run.

Each record represents a single observing block and shows its name, observation run to which it belongs, description, observing semester, start date and time, end time, observed objects (targets), observation mode (*Photometry*, *Spectroscopy*, or *Imaging*), filters, configuration, exposure time of the observation, seeing value, weather conditions, and comments.

Observing Blocks table

NAME	OBSERVING RUN	DESCRIPTION	OBSERVING SEMESTER	START TIME	END TIME	TARGET	OBSERVATION MODE	FILTERS	CONFIGURATION	...
GT4-23ITP	GT4-23ITP	NGC 6822, IFS with MEGARA (23ITP)	2023a	Feb. 18, 2023, midnight	12:10 a.m.		photometry	LR-U		

Figure II.14. *Observing Blocks* table.

6.8 Targets

The *target* table contains information about the observable astronomical objects. Targets are displayed in ascending order of right ascension.

Each record represents a single target and shows its image (if available), name, observing blocks where it has been observed, type (*Galaxy*, *Calibration*, or *Other*), right ascension, declination, apparent magnitude, redshift, angular size, visibility semester, comments, path to store associated files and a link to a website with additional information.

Targets table

IMAGE	NAME	OBSERVING BLOCK	TYPE	RA (HH-MM-SS)	DEC (DD-MM-SS)	MAG <i>r</i> (AB)	REDSHIFT VALUE	REDSHIFT ERROR	SIZE (arcsec)	VISIBILITY SEMESTER	COMMENTS	DATA FILES PATH	WEBSITE
	UGC4483		galaxy	08:37:02.9721	+69:46:32.800	0.000519	0.000001	84.0	None			UGC4483/datafiles	↗
	M81DwarfB	block-w1 - WEAVE (2025-10-25)	galaxy	10:05:30.1495	+70:21:50.866	14.7889	0.001159	0.000002	78.0	None	Low Surface Brightness Galaxy	M81DwarfB/datafiles	↗
	J1105+4444		galaxy	11:05:08.1192	+44:44:47.241		0.021541	-0.99999117	40.33	None	Object MRK 0162	J11054444/datafiles	↗
	Mrk1486	GT4-23ITP - MEGARA (2023-02-18)	galaxy	13:59:50.9698	+57:26:23.233	16.809	0.033829	0.00000449	13.38	None	HII Galaxy Wolf-Rayet Galaxy	Mrk1486/datafiles	↗

Figure II.15. *Targets* table.

7. Administration Panel

The *administration panel* offers a centralized interface to maintain the database, and manage user accounts and permissions.

The header menu presents the following options for the **Administrator** user:

- **View Site:** returns to the platform's main page (see [Chapter 3. Navigating the Home Page](#)).
- **Change Password:** allows the **Administrator** to update their password.
- **Log out:** closes the current session (see [Chapter 8. Logging Out](#)).



Figure II.16. Administration's panel menu options.

7.1 Panel Overview

The administration panel is organized into two main sections.

Authentication and Authorization section contains the *Groups* and *Users* categories, to manage user accounts, assign users to groups, and configure permissions.

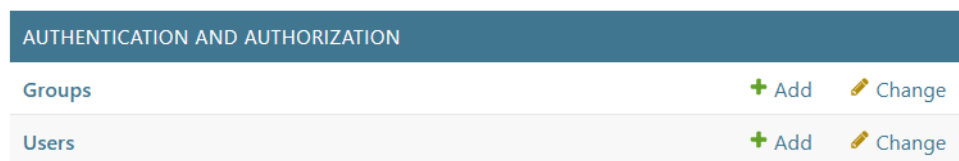


Figure II.17. Authentication and Authorization administration.

Database section defines a category for each table of the *dwarfs4MOSAIC* platform, allowing to view, add, modify, or delete records as required.

DATABASE		
Instruments	+ Add	Change
Observatories	+ Add	Change
Observing Blocks	+ Add	Change
Observing Runs	+ Add	Change
Researchers	+ Add	Change
Targets	+ Add	Change
Telescopes	+ Add	Change

Figure II.18. Database administration.

When selecting a category in the administration panel, a list of existing objects of that type is displayed, allowing you to view, edit, or delete them as needed.

Select Instrument to change

Action: 0 of 5 selected

☐	INSTRUMENT
☐	IO:O
☐	MEGARA
☐	SIFS
☐	WEAVE
☐	WFC

5 Instruments

Figure II.19. List of instruments.

All categories in the administration panel have the same basic actions.

Importing a set of objects

For each category, an *Import CSV* button is available.



Figure II.20. Import CSV button.

Clicking this button opens a dedicated view to import objects of the selected category. The **Select File** button opens a file browser to choose a CSV file to import. This file must include all fields corresponding to the category and contain a header row with the database column names.

Import observatories from CSV

Important notice:
The CSV is downloaded *as-is* without any data validation.

IDs are not preserved and cannot be managed through this file.
After downloading, you must manually update the objects to define their relationships with other tables.

Additionally, any **associated files must be uploaded manually** after the import.

CSV file: Select File **Selected file:** dwarfs4MOSAIC_tbl_observatory.csv

Import Cancel

Figure II.21. Import CSV view.

 The CSV is imported as-is, without any data validation. Object IDs in the CSV are not preserved; the platform automatically assigns new IDs upon import. After importing, you must manually update the objects to establish their relationships with other tables, and upload any associated files separately.

Adding an object

Open the form to create a new object:

- By clicking **Add+** next to the category name in the main panel.
- By selecting a category and then clicking **Add [object name]+**.
- By clicking **Save** and **add another** after creating an object, which immediately opens a new form for the same type of object.



Figure II.22. Add a new object.

Fill in the required fields (shown in **bold**) and any optional fields as needed, and click **Save** to create the object. After saving, the new object will appear in the list of objects of that category, and you can edit it at any time to view it, update it or delete it if it is no longer needed.

Editing an object

Select a category and click on the object you wish to modify from the list. Update the fields as needed and click **Save** to apply the changes.

Deleting an object

Select a category, click on the object from the list to open its form, and click **Delete**. You will be asked to confirm before the deletion is applied.

It is also possible to delete multiple objects at once by selecting several entries in the list and choosing the **Delete selected** action from the dropdown menu.

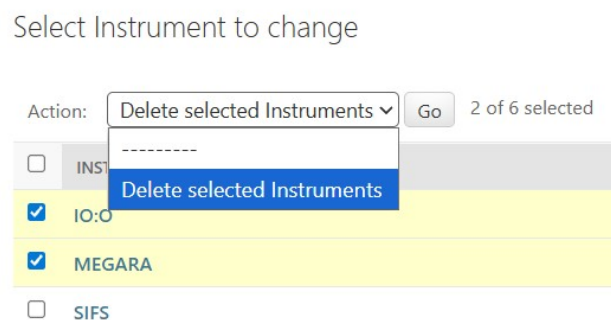


Figure II.23. Delete existing object/s.

7.2 Database

This section describes the database categories in an order that reflects the typical workflow for creating and linking objects (see [Figure II.3. Database structure and relationships between tables](#)). Presenting them in this sequence helps clarify the relationships between different types of entries and ensures a logical understanding of how data is organized within the platform.

7.2.1 Observatories

Observatories represent the locations where telescopes and instruments are installed.

To register a new observatory, click *Add Observatory* and provide the required information in the following fields:

Name: name of the observatory.

Add Observatory

Name:

Figure II.24. Observatory name.

General Information

Location: city, region or country where the observatory is located.

Website: official website of the observatory (if available).

General Information	
Location:	La Palma (Spain)
Website:	https://www.iac.es/es/observatorios-de-canarias/observatorio-del-roque-de-los-muchachos

Figure II.25. Observatory general information.

Coordinates

Longitude: geographical longitude of the observatory, expressed in degrees, minutes, seconds (may include decimal values) and the direction (East (**E**) or West (**W**)).

Latitude: geographical latitude of the observatory, expressed in degrees, minutes, seconds (may include decimal values) and the direction (North (**N**) or South (**S**)).

Altitude: geographical height above sea level, expressed in meters.

Coordinates	
Longitude:	17:53:30W DDD:MM:SS[.sss][E/W]
Latitude:	28:45:22N DD:MM:SS[.sss][N/S]
Altitude:	2396.0 meters

Figure II.26. Observatory coordinates.

Maintaining accurate observatory data is essential, as telescopes and observing blocks are linked to these locations, and the relationships are used throughout the platform to manage observations efficiently.

7.2.2 Telescopes

Telescopes are the instruments installed at observatories to carry out astronomical observations.

To register a new telescope, click *Add Telescope* and provide the required information in the following fields:

Name: name of the telescope.

Add Telescope

Name:

Figure II.27. Telescope name.

General Information

Description: short description of the telescope.

Institutional Owner: institutional owner of the telescope.

Observatory: observatory where the telescope is located.

Website: official website of the telescope (if available).

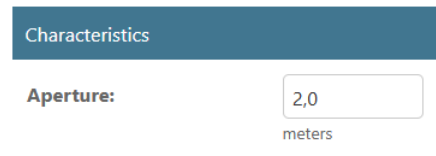
Status: current operational status of the telescope (*Unknown*, *Operational*, *Inoperative*, *Under Maintenance*).

General Information	
Description:	Liverpool Telescope
Institutional Owner:	Liverpool John Moores University
Observatory:	Roque de los Muchachos    
Website:	https://telescope.livjm.ac.uk/
Status:	Operational ▼

Figure II.28. Telescope general information.

Characteristics

Aperture: aperture of the telescope, expressed in meters.



The screenshot shows a dark blue header bar with the text 'Characteristics'. Below it, the label 'Aperture:' is followed by a text input field containing the value '2,0'. Below the input field, the unit 'meters' is displayed.

Figure II.29. Telescope coordinates.

Accurate telescope data is important because instruments, observing blocks, and targets are linked to specific telescopes. Ensuring these relationships are correctly established allows proper management and planning of observations across the platform.

7.2.3 Instruments

Instruments are the devices attached to telescopes used to acquire observational data, such as cameras, spectrographs, or photometers.

To register a new instrument, click *Add Instrument* and provide the required information in the following fields:

Name: name of the instrument.

Add Instrument



The screenshot shows a form with the label 'Name:' followed by a text input field containing the value 'WEAVE'.

Figure II.30. Instrument name.

General Information

Description: short description of the instrument.

Telescope: telescope where the instrument is installed.

Website: official website of the instrument (if available).

Status: current operational status of the instrument (*Unknown*, *Operational*, *Inoperative*, *Under Maintenance*).

General Information	
Description:	WHT Enhanced Area Velocity Explorer
Telescope:	WHT    
Website:	https://www.ing.iac.es//astronomy/instruments/weave/weaveinst.html
Status:	Operational 

Figure II.31. Instrument general information.

Additional Data

Filters: filters or filter set used by the instrument. Specify all available filters or bands if applicable.

Configuration: optical or mechanical configuration of the instrument.

Additional Data	
Filters:	
	Comma-separated
Configurations:	LR-blue, LR-red 
	Comma-separated

Figure II.32. Instrument additional data.

Correctly registering instruments ensures that observing blocks and targets are properly associated with the hardware used, which is essential for data management and planning of observations.

7.2.4 Observing Runs

Observing runs represent the periods of scheduled observations conducted at a given observatory with specific telescopes and instruments.

To register a new observing run, click *Add Observing Run* and provide the required information in the following fields:

Name: name of the observing run.

Add Observing Run

Name:

GTC4-23ITP

Figure II.33. Observing run name.

General Information

Description: description of the observing run.

Instrument: instrument used during the observing run.

Start Date: start date of the observing run.

End Date: planned end date of the observing run.

General Information

Description:

The observing run will be conducted at the 10.4m Gran Telescopio Canarias (GTC) in the framework of the International Time Programme 2023 (23ITP).

It consists of a series of observing blocks targeting selected galaxies, each designed to deliver integral field spectroscopy datacubes for subsequent analysis.

Instrument:

MEGARA



Start Date:

2023-02-18

Today | 

Note: You are 2 hours ahead of server time.

End Date:

2023-02-19

Today | 

Note: You are 2 hours ahead of server time.

Figure II.34. Observing run general information.

Participants

Researchers: researchers participating in the observing run.

The 'Participants' interface is divided into two main sections: 'Available researchers' and 'Chosen researchers'. Both sections have a search bar labeled 'Filter'. The 'Available researchers' list includes: Cristina Cabello González, Christopher J. Conselice, Davor Krajnovic, Jorge Iglesias Páramo, Luca Costantin, Matthew Hayes, Maritza Lara López, Mathieu Puech, Nicolas Laporte, Roser Pello, and Rubén Sánchez Janssen. The 'Chosen researchers' list includes: África Castillo Morales, Armando Gil de Paz, Carolina Kehrig Martin dos Santos, and José Manuel Vilchez Medina. Below the lists are buttons for 'Choose all' and 'Remove all'. A note at the bottom states: 'Researchers who participated in the observing run. Hold down "Control", or "Command" on a Mac, to select more than one.'

Figure II.35. Observing run participants.

Additional Data

Comments: any additional information concerning the observing run.

The 'Additional Data' interface features a 'Comments' section with a text input area. The comment entered is: 'The run was executed over several nights under photometric conditions with typical seeing below 1.0".

Figure II.36. Observing run additional data.

Properly registering observing runs is crucial for organizing observing blocks, associating data files, and maintaining accurate records of the observations carried out.

7.2.5 Observing Blocks

Observing blocks are subdivisions of an observing run that define specific targets, instruments, and observation parameters for a given session.

To register a new observing block, click *Add Observing Block* and provide the required information in the following fields:

Name: name of the observing block.

Add Observing Block

Name:

Figure II.37. Observing block name.

General Information

Observing run: observing run where the block is executed.

Description: description of the block.

Observing semester: semester in which the block was executed.

Start Time: start date and time of the block.

End Time: planned end time of the block.

General Information	
Observing Run:	GTC4-23ITP    
Description:	NGC 6822, IFS with MEGARA (23ITP). 
Observing semester:	2023a
Start Time:	Date: 2023-02-18 Today  Time: 00:00:00 Now  Note: You are 2 hours ahead of server time.
End Time:	00:10:00 Now  Note: You are 2 hours ahead of server time.

Figure II.38. Observing block general information.

Observation Information

Observation mode: observation mode used during the observing block (*Photometry*, *Spectroscopy*, or *Imaging*).

Instrument filter: instrument filter used during the observations.

Instrument configuration: instrument configuration used during the observations.

Exposure Time: exposure time per observation, expressed in seconds.

Seeing: seeing value during the observations, expressed in arcseconds.

Weather Conditions: weather conditions during the observing block.

Target: targets observed in the block.

The screenshot shows a web form titled "Observation Information". It contains several input fields and sections:

- Observation Mode:** A dropdown menu set to "Photometry".
- Instrument filter:** A dropdown menu set to "LR-U".
- Instrument configuration:** A dropdown menu set to "-----".
- Exposure Time:** A text input field with "1200" and a label "seconds" below it.
- Seeing:** A text input field with "1.0" and a label "arcsec" below it.
- Weather Conditions:** A text area containing: "Sky transparency: Photometric (no clouds)", "Moon illumination: < 30% (dark/grey time)", "Airmass: ≤ 1.3", "Humidity: ≤ 70%", and "Wind speed: ≤ 10 m/s".
- Target:** Two side-by-side panels. The left panel, titled "Available target", lists "IC 1613", "Leo I", "Pegasus Dwarf", and "Sextans A". The right panel, titled "Chosen target", lists "NGC 6822". Both panels have a "Filter" search bar and a "Choose all" / "Remove all" button at the bottom.

Below the target panels, there is a note: "Targets that belong to the block. Hold down 'Control', or 'Command' on a Mac, to select more than one."

Figure II.39. Observing block observation information.

Additional Data

Comments: any additional information concerning the observing block.

The screenshot shows a web form titled "Additional Data". It contains a text area labeled "Comments:" with the following text: "Observing block 0009: integral field spectroscopy of the dwarf galaxy NGC 6822 with GTC/MEGARA as part of the 23ITP run."

Figure II.40. Observing block additional data.

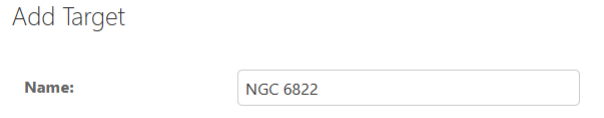
Proper configuration of observing blocks ensures that data files are correctly associated with their targets and runs, and that access permissions can be assigned effectively to researchers and groups.

7.2.6 Targets

Targets are the astronomical objects of study in the project.

Creating a new *target* is a two-step process. First, click *Add Target* and provide the required information in the following fields:

Name: name of the target (must be unique).



Add Target

Name:

Figure II.41. Target name.

General Information

Type: target type (*Galaxy*, *Calibration*, or *Other*).

Website: link to an external astronomical database or information page for the object.

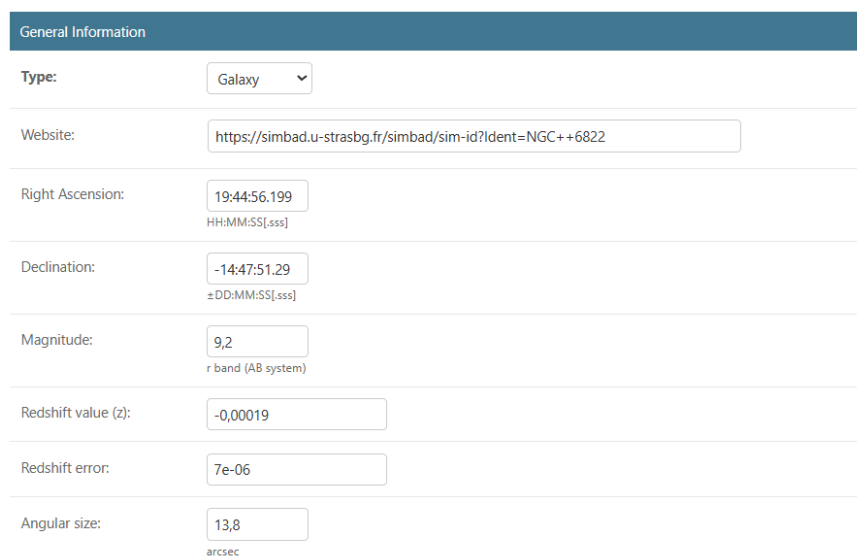
Right Ascension: right ascension of the target, expressed in hours, minutes, and seconds.

Declination: declination of the target, expressed in degrees, arcminutes, and arcseconds, with positive values for northern and negative for southern declinations.

Magnitude: apparent magnitude of the object in the *r* band (AB system).

Redshift: redshift of the object with error.

Size: apparent angular size of the object, given in arcseconds.



General Information

Type:

Website:

Right Ascension:
HH:MM:SS[.sss]

Declination:
±DD:MM:SS[.sss]

Magnitude:
r band (AB system)

Redshift value (z):

Redshift error:

Angular size:
arcsec

Figure II.42. Target general information.

Additional Data

Visibility semester: semester in which the target is visible.

Comments: any additional information concerning the target.

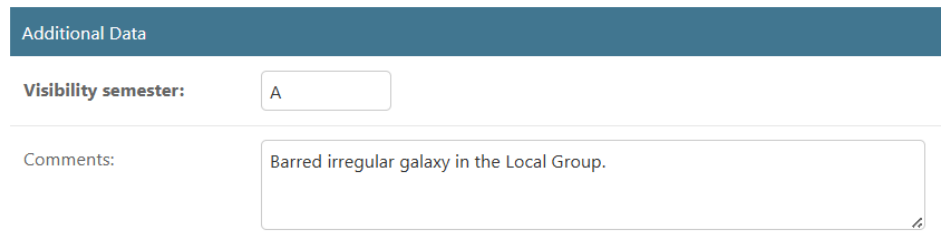


Figure II.43. Target additional data.

After saving, you will be redirected to the target details page. Here, fill in the following fields:

Upload Files

Image: image file for the target. Clicking the **Select File** button opens a file browser window to choose the image from your computer.

Data files: target data files. Clicking the **Select Files** button opens a file browser window to select one or multiple files to upload.

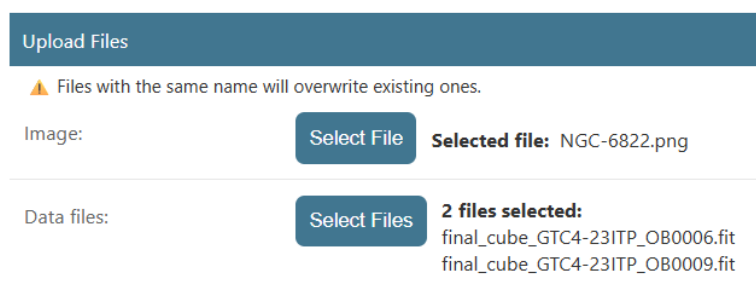


Figure II.44. Target files upload.

Delete Files

Delete image: Check this box to remove the currently associated image file from the target.

Delete files: Select one or more entries from the list to delete the corresponding data files.

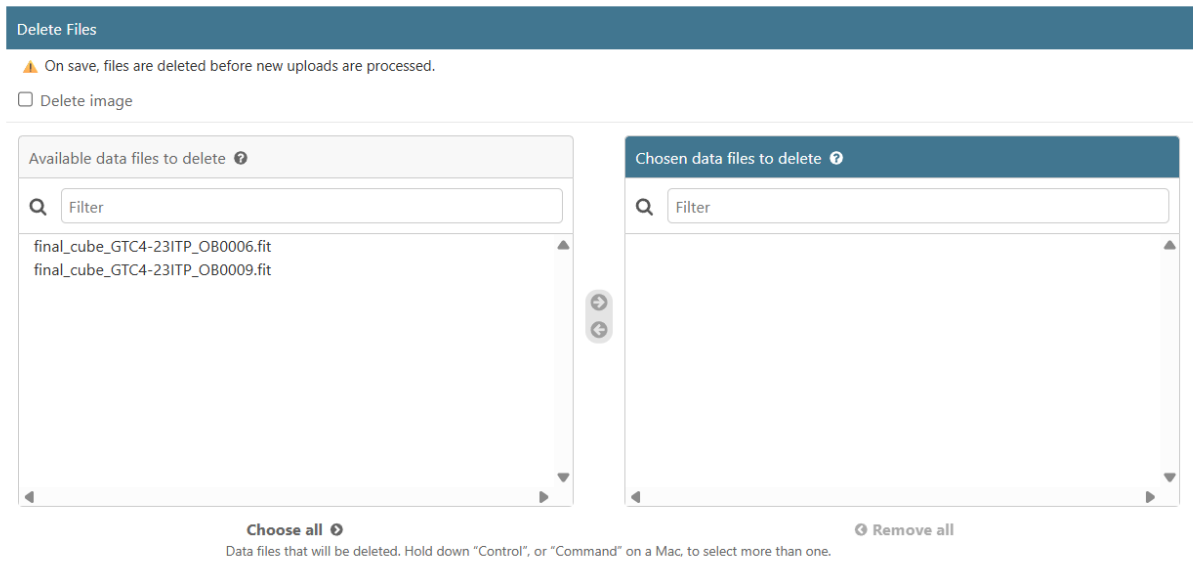


Figure II.45. Target files deletion.

Correct configuration of targets allows for accurate organization of observations, data management, and control of user access permissions for each object.

7.3 Authentication and Authorization

The platform defines different permissions to manage access to its features and data.

The **Administrator** has full access to all platform functions, including user and group management, database maintenance, and content editing.

The remaining researchers can have one of two roles:

- **Core Team:** full access to all targets and observing blocks.
- **Collaborator:** restricted to the targets and observing blocks assigned to their group; any blocks explicitly denied remain inaccessible.

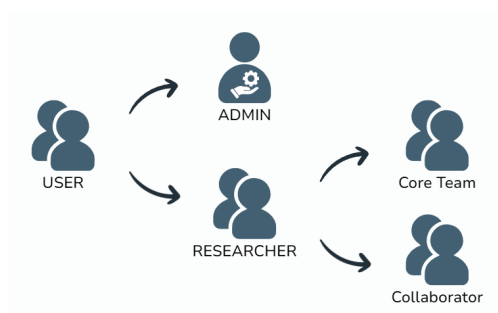


Figure II.46. Users profiles and roles.

7.3.1 User/Researcher Management

Every member involved in the project is represented by two linked records: a **User** and a **Researcher**.

The *User* account manages the basic login credentials and general access settings, such as password and group membership. The *Researcher* record extends this information by adding specific details about the person as a scientist, such as role, PhD status, and permissions to access observing blocks.

When creating a new member, the process always begins with the *User* account. Once the account is saved, the corresponding *Researcher* information can be added and configured.

Creating a new user/researcher

Create account

First, *add user* or *add researcher* from the corresponding category. In both cases, the *Add User* form will be opened.

In this initial form, you must enter the username for the new account and set a password. The password can be typed manually twice for confirmation, or it can be generated automatically by the system. If generated, the password will be displayed so that you can share it with the new user by email.

Add user

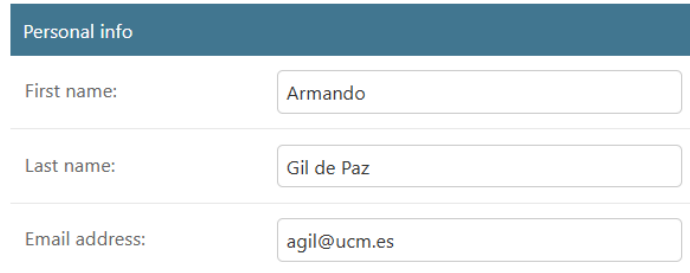
The screenshot shows a web form titled "Add user". It contains the following elements:

- Username:** A text input field containing "agil". Below it, a note reads: "Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only."
- Password-based authentication:** Two radio buttons are present: "Enabled" (which is selected) and "Disabled". Below them, a note reads: "Whether the user will be able to authenticate using a password or not. If disabled, they may still be able to authenticate using other backends, such as Single Sign-On or LDAP."
- Generate password:** A dark blue button with white text.
- Password:** A text input field containing "vb~wV&WiU[t5". Below it, four validation messages are listed:
 - Your password can't be too similar to your other personal information.
 - Your password must contain at least 8 characters.
 - Your password can't be a commonly used password.
 - Your password can't be entirely numeric.
- Password confirmation:** A text input field containing "vb~wV&WiU[t5". Below it, a note reads: "Enter the same password as before, for verification."

Figure II.47. Creating a new user.

Add user information

After saving, you will be redirected to the user details page. In this second step, you can complete the personal information fields (first name, last name, and email).

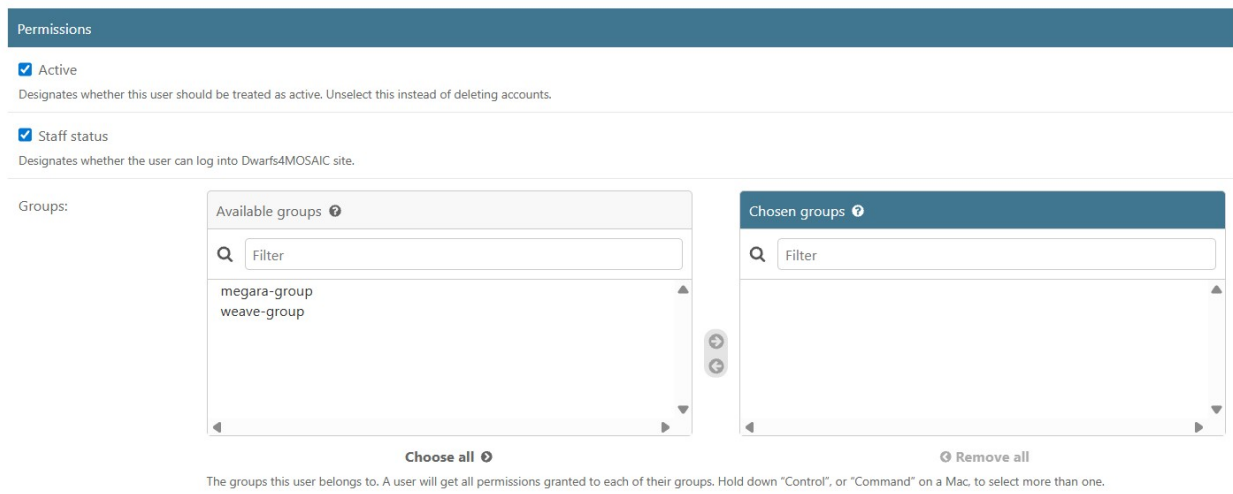


A screenshot of a web form titled "Personal info" with a dark blue header. Below the header are three input fields. The first field is labeled "First name:" and contains the text "Armando". The second field is labeled "Last name:" and contains the text "Gil de Paz". The third field is labeled "Email address:" and contains the text "agil@ucm.es".

Personal info	
First name:	Armando
Last name:	Gil de Paz
Email address:	agil@ucm.es

Figure II.48. User personal information.

Also, you can assign staff status if needed, and manage groups and permissions to control access to targets, blocks, and administrative features. To allow the user to access the platform, the **Staff status** option must be checked.



A screenshot of a web form titled "Permissions" with a dark blue header. Below the header are two checkboxes. The first checkbox is labeled "Active" and is checked; below it is the text "Designates whether this user should be treated as active. Unselect this instead of deleting accounts." The second checkbox is labeled "Staff status" and is checked; below it is the text "Designates whether the user can log into Dwarfs4MOSAIC site." Below these checkboxes is a section labeled "Groups:". It contains two side-by-side list boxes. The left list box is titled "Available groups" and contains a search bar with the text "Filter" and two items: "megara-group" and "weave-group". Below the list box is a button labeled "Choose all". The right list box is titled "Chosen groups" and is currently empty. Below the list box is a button labeled "Remove all". At the bottom of the "Groups:" section is a note: "The groups this user belongs to. A user will get all permissions granted to each of their groups. Hold down 'Control', or 'Command' on a Mac, to select more than one."

Permissions

☒ Active
Designates whether this user should be treated as active. Unselect this instead of deleting accounts.

☒ Staff status
Designates whether the user can log into Dwarfs4MOSAIC site.

Groups:

Available groups

megara-group
weave-group

Choose all

Chosen groups

Remove all

The groups this user belongs to. A user will get all permissions granted to each of their groups. Hold down "Control", or "Command" on a Mac, to select more than one.

Figure II.49. User permissions.

This page also displays additional information automatically managed by the system, such as the dates when the user account was created (**Date joined**) and the last time the user logged in (**Last login**).

Important dates

Last login:

Date: 2025-08-23 Today | 📅

Time: 10:47:41 Now | ⌚

Note: You are 2 hours ahead of server time.

Date joined:

Date: 2025-08-23 Today | 📅

Time: 10:29:14 Now | ⌚

Note: You are 2 hours ahead of server time.

Figure II.50. User dates.

Add researcher information

Finally, the researcher information can be added either directly from the user page by clicking **Open Researcher**, or by selecting an existing researcher from the *Researchers* category.

In this form, you can complete all fields related to the researcher's role, PhD status, and permissions for observing blocks. This ensures that the account is fully linked to its corresponding researcher profile and that all relevant access rights are properly assigned.

You can access to the corresponding user account by clicking on the username link, allowing you to view or edit the account information directly.

Use the Role field to assign *Core Team* or *Collaborator* status. Core Team members have full access to all targets and observing blocks, while Collaborators are restricted to the targets and blocks assigned to their group.

Armando Gil de Paz

Username: agil

Role:

Core Team ▼

Figure II.51. Researcher role.

In the *General Information* section you can indicate whether the researcher is a PhD, the institutional affiliation of the researcher and any additional informational about the researcher.

General Information

☒ Is PhD

Institution:

Comments:

Figure II.52. Researcher general information.

The *Authorization* section shows the *Denied blocks* field, to select which observing blocks the researcher does not have permission to access.

Authorization

Denied blocks:

Available denied blocks

Filter

GTC4-23ITP

Choose all

Chosen denied blocks

Filter

Remove all

The user does not have authorized access to these blocks (even if they belong to a group that does have authorized access). Hold down "Control", or "Command" on a Mac, to select more than one.

Figure II.53. Researcher Authorization.

Deleting a user/researcher

When a user account is deleted, the login credentials are removed, but the researcher record remains in the database. This ensures that the researcher's participation and history in past observing campaigns are preserved. However, once the associated user account is deleted, the researcher record becomes read-only and can no longer be edited.

⚠ Conversely, if a researcher record is deleted, both the researcher and the linked user account are permanently removed.

⚠ The **Administrator** account cannot be deleted for security reasons.

7.3.2 Groups Management

Groups are used to organize users according to their roles and responsibilities within the project. By assigning users to a group, you can efficiently manage permissions and control access to targets, observing blocks, and administrative features.

Creating a new *group* is a two-step process. First, *add group* and enter the group name. After saving, you will be redirected to the group details page.

Add group

Name:

Figure II.54. Group name.

In a second step, the *Authorization* section appears, where you can assign the group access to specific observing blocks. This specifies which observing blocks the users in the group are allowed to access or modify.

Authorization

Available Observing Blocks ⓘ

Q Filter

WEAVE NGC 6822 - 2025B-OB001 - WEAVE (2025-09-01)

Choose all ⓘ

Chosen Observing Blocks ⓘ

Q Filter

GTC4-23ITP - MEGARA (2023-02-18)

Remove all ⓘ

Authorized blocks for users belonging to the group. Hold down "Control", or "Command" on a Mac, to select more than one.

Figure II.55. Group allowed blocks.

Users can then be assigned to the group when creating or editing their accounts.

8. Logging Out

For instructions on logging out, the Administrator should refer to *Chapter 6. Logging Out* in *Part I. User's Guide* .

9. Support

For support-related issues, the Administrator should refer to *Chapter 7. Support* in *Part I. User's Guide* .